

Potential course project ideas

BIS 527: Introduction to Health Data Science

Fall 2026

These are starting points, not a menu you have to choose from. Your own idea is welcome, and a good project usually comes from a question you already care about. Teams are 4 to 5 students. Where a contact is listed, write to them early: they know the data and can tell you quickly whether a question is answerable. Where approval is needed, start that request before you write the proposal, because it will not arrive in time otherwise.

Fertility and maternal mortality

Data. <https://www.census.gov/programs-surveys/acs/data.html>, <https://www.bls.gov/tus/database.htm>

Over the last 25 years there has been a global effort to reduce maternal mortality across countries, reflected in both the Millennium Development Goals and the Sustainable Development Goals. Over the same period many countries have gone through a demographic transition toward substantially lower fertility and later age at first birth. This project separates the effects of population growth, aging, and age-specific fertility on changes in maternal mortality across countries. It then asks whether stagnation in some countries can be explained by high fertility or by other population characteristics. The project uses publicly available data from the WHO and the UN.

Note. Contact: Emily Johnson (SDU)

Medicaid expansion and state health spending

Data. <https://www.census.gov/programs-surveys/acs/data.html>, <https://www.bls.gov/tus/database.htm>, <https://civhc.org/standard-data-sets/>

Medicaid expansion under the Affordable Care Act extended coverage to millions of low-income adults, and a large literature documents that the newly eligible used more care. Much less is known about who ends up paying for that care once expansion takes effect. This project asks how expansion reshaped the full payer mix of US healthcare: whether new Medicaid spending displaced out-of-pocket and private spending, whether there were spillovers onto Medicare, and whether these patterns differ across care settings such as ambulatory, emergency, inpatient, nursing facility, and pharmacy care. The analysis uses state-level, age- and sex-standardized estimates of healthcare spending and utilization from the Institute for Health Metrics and Evaluation's Disease Expenditure project (DEX 2.0), which covers five types of care and four payers (Medicaid, Medicare, private insurance, and out-of-pocket) annually from 2010 to 2019. These are combined with state expansion timing from KFF and demographic and economic covariates from the CPS.

Note. Contact: Emily Johnson (SDU)

Vector-borne disease dataset generation

Data. State arbovirus surveillance pages, for example Connecticut: https://portal.ct.gov/caes/mosquito-testing/introductory/state-of-connecticut-mosquito-trapping-and-arbovirus-testing-program?language=en_US

Many states publish vector-borne disease and vector surveillance data, but each one publishes in its own non-standard format, so the results cannot easily be compared across states. This project develops scrapers that pull all of this

information into one place in a single standard format. Most of the work lies in deciding what a shared format should look like when states report different species, different trapping effort, and different testing practices, and then documenting the choices made to reconcile them. The deliverable is a usable dataset together with the documentation that makes it trustworthy.

Note. Contact: Dan Weinberger (PopHive)

Association between AI use and mental health outcomes

AI and digital media have been shown anecdotally to affect people's lives in both positive and negative ways. An open question for the coming years is whether and how exposure to them affects individual mental health. Individual-level AI usage data may be hard to obtain, but aggregated state-level or county-level usage patterns can be compared against mental health outcomes over time. Questions worth asking include whether any association survives once local economic conditions are accounted for, whether it differs by age group, and what an aggregate pattern can and cannot tell us about individual people.

Note. Contact: Harsh Parikh and Emily Johnson

New data collection opportunity: how are BIS 527 students using AI

How are BIS 527 students using AI, and how does it affect their learning outcomes? The goal is to work out what an optimal AI use strategy looks like if the aim is to maximize learning. This is the only idea on the list where the team designs the instrument and collects the data itself, so the decisions about what to ask, whom to ask, and how to measure learning are the project rather than a preliminary to it. Angles worth considering include which tasks students hand to AI and which they keep, whether declared AI use tracks with performance on homework and quizzes, and how students decide whether an AI answer is right.

Note. May require IRB approval. Check with Harsh beforehand.

Association between air quality and respiratory illness

Global carbon emissions have recently been on the rise and are expected to grow further as AI data centers are built out. A great deal of work has already been done on the link between long-term PM2.5 exposure and mortality, and on wildfire-specific PM2.5, so the open questions are narrower. Two worth asking are which subgroups are most vulnerable, particularly children, and how the effects vary geographically. The second matters because most research in this area has been conducted in North America, which leaves other regions of the world underrepresented.

Ibrutinib treatment discontinuation patterns

Data. Yale Open Data Access Project (YODA): ibrutinib (IMBRUVICA) trials

Ibrutinib has known tolerability problems, including atrial fibrillation, bleeding, and arthralgia, that lead patients to stop treatment early for reasons other than disease progression. This project characterizes the reasons for and the timing of discontinuation across trials, and models which baseline factors predict stopping early rather than completing therapy.

Note. Requires permission from the trial PI, so budget time for approval. For advice on how to frame your research question, contact Yukta.

Pulsing technique detection from audio signal

Data. YODA: ECHELON FLEX GST staple line videographic study

The study states that audio was collected in order to identify the surgeon's pulsing technique from motor noise during firing. This project builds a signal processing pipeline, using spectral analysis and, if enough labeled examples exist, a simple classifier, to detect pulsing against continuous firing from the audio track. The resulting classifications are then checked against the study's own technique labels.

Note. Requires permission from the trial PI, so budget time for approval.

Primary efficacy replication: potassium oxalate rinse vs. placebo

Data. YODA: Potassium Oxalate 1.4% trial

This project reproduces the trial's primary analysis: the proportion of participants achieving at least a 30% reduction in cold air VAS score at Week 4, comparing the KOX rinse to placebo. The analysis itself is a straightforward two-proportion comparison by chi-square or regression.

Note. Requires permission from the trial PI, so budget time for approval. For help, contact Yukta.

Crime, time and Yale Alerts

Data. Yale Alerts data

Yale Alerts notify the campus community about crimes reported around the university, and the accumulated alerts form a record of where and when incidents occur. This project maps that record to find hotspots around campus and to describe how risk shifts across the hours of the day, the days of the week, and the academic calendar. It then turns that picture into something a student could act on: given a starting point and a destination, which route carries the least exposure, and how much does the answer change between late afternoon and midnight? A team taking this on should think carefully about what the alerts actually capture, since an alert records an incident that was reported and deemed worth notifying rather than all crime that occurred.

Note. Contact: Yi Ting (PHMU, Yale SPH)

Questions about scope or feasibility: bring them to office hours, or email harsh.parikh@yale.edu.